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Week 9 Quiz

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Week 9 Lectures

Week 9 Quiz

Quiz: Week 9 Quiz

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1. The name of the convolutional neural network derives from the fact that the connections between layers are made in localised groups.

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1. The name of the convolutional neural network derives from the fact that the connections between layers are made in localised groups.

1 / 1 point

☒ True

☐ False

👍 Correct

That is correct. The localised groups form a convolution-like operation

2. The ReLU activation function in a convolutional neural network is used instead of the sigmoid and other functions used in the multi-layer Perceptron because (choose one):

1 / 1 point

☐ It generates more accurate results

☒ It allows faster convergence during training

☐ It improves the multi-class capabilities of the algorithm

👍 Correct

That is correct.

3. Accounting for spatial context when classifying an image is important because:

1 / 1 point

☐ Most scenes contain well-defined objects

☐ Spatial properties are more important than spectral properties

☒ Neighbouring pixels are likely to be from the same class.

👍 Correct

That is correct. Most environments exhibit this property (natural, agricultural, urban, water bodies, etc.)

4. What is the reason for including a pooling layer in a convolutional neural network?

1 / 1 point

☒ It further reduces the number of connections and helps imbed spatial context

☐ Convolutional layers require a pooling layer as their inter-connection

☐ It removes the need for an activation function in the preceding convolutional layer

👍 Correct

That is correct.

5. The softmax function often used with convolutional neural networks

1 / 1 point

☒ Produces a set of probabilities from the outputs

☐ Avoids the spread of the output values covering too great a range

☐ Makes it easier to see which is the preferred output

👍 Correct

That is correct.

6. The stride operation in a convolutional neural network (choose one)

1 / 1 point

☐ Has no effect of the spatial properties being learnt

☒ Allows the number of connections and thus unknowns to be reduced

☐ Must always be smaller than 3

👍 Correct

That is correct and is the reason stride is sometimes built into a CNN.

7. Feature (e.g. band) reduction is often applied to hyperspectral imagery before being fed into a convolutional neural network because (choose one)

1 / 1 point

☐ CNNs can't cope with large data sets

☐ We never need to use all the bands anyway

☒ The number of unknowns found during training is reduced

👍 Correct

That is correct. It simplifies training and can lead to a more accurate classifier—one which generalises well.

8. Compared with a fully connected neural network, convolutional neural nets (choose one):

1 / 1 point

☒ Can have more hidden layers because of the fewer connections between nodes

☐ Always perform better when spatial context is a consideration

☐ Can have fewer output nodes

👍 Correct

That is correct, because the number of unknowns to be found during training is reduced.

9. Operating several CNN pathways in parallel is designed to (choose one)

1 / 1 point

☐ Speed up training

☐ Speed up the classification step

☒ Respond to spatial features at different scales

👍 Correct

That is correct and is often the reason several parallel paths (filters) is used.

10. Flattening (choose one answer):

1 / 1 point

☒ is an operation that turns an array into a column vector

☐ Reduces the dynamic range of the values in a measurement vector

☐ Is a method for combining two or more arrays

👍 Correct

That is correct

1/1

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